HR Employee Attrition — EDA Practice Task Sheet

Practice dataset: HR_Employee_Attrition_EDA_Practice.csv
Difficulty: Beginner → Intermediate
Original dataset: 1,470 rows | **Practice dataset:** 1,505 rows
No answers are included. Discover the data-quality issues yourself.

Part 1 — Dataset Understanding

Q1. Load the CSV into a DataFrame named `df`.

Q2. Find the number of rows and columns, and display the first and last five rows.

Q3. Display all column names and their data types. Classify columns as numerical, categorical, or identifier-like.

Q4. Use `info()` and identify columns that may need special attention.

Part 2 — Data Quality Investigation

Q5. Find the number and percentage of missing values in every column.

Q6. Check for duplicate rows. How many duplicates are present?

Q7. Investigate unique values in categorical columns such as Attrition, BusinessTravel, Department, Gender, JobRole, MaritalStatus, OverTime, and EducationField.

Q8. Look for inconsistent capitalization, leading/trailing spaces, and unexpected categories.

Q9. Check numerical columns for impossible or suspicious values. Do not clean them yet—document what you find.

Part 3 — Univariate Analysis

Q10. Use descriptive statistics for Age, DailyRate, DistanceFromHome, HourlyRate, MonthlyIncome, MonthlyRate, TotalWorkingYears, YearsAtCompany, and YearsWithCurrManager.

Q11. Create histograms for at least five important numerical variables.

Q12. Create boxplots for Age, MonthlyIncome, DistanceFromHome, and TotalWorkingYears.

Q13. Analyze the distribution of Attrition, Gender, Department, JobRole, OverTime, and BusinessTravel.

Part 4 — Attrition Analysis

Q14. Calculate the total number and percentage of employees who left versus stayed.

Q15. Compare attrition rate by Department.

Q16. Compare attrition rate by JobRole.

Q17. Compare attrition rate by OverTime.

Q18. Compare attrition rate by BusinessTravel.

Q19. Compare attrition rate by MaritalStatus and Gender.

Q20. Identify which groups have noticeably higher or lower observed attrition rates.

Part 5 — Numerical Relationships

Q21. Compare MonthlyIncome between employees who left and stayed.

Q22. Compare Age, DistanceFromHome, TotalWorkingYears, YearsAtCompany, and JobSatisfaction by Attrition.

Q23. Create suitable visualizations for at least four numerical variables against Attrition.

Q24. Investigate whether YearsAtCompany and YearsInCurrentRole show an apparent relationship with attrition.

Part 6 — Satisfaction & Work Conditions

Q25. Analyze EnvironmentSatisfaction, JobInvolvement, JobSatisfaction, RelationshipSatisfaction, and WorkLifeBalance.

Q26. Calculate attrition rate for each satisfaction level.

Q27. Investigate the relationship between OverTime and JobSatisfaction.

Q28. Investigate whether StockOptionLevel appears related to Attrition.

Part 7 — Correlation Analysis

Q29. Create a correlation matrix for numerical variables.

Q30. Visualize the correlation matrix using a heatmap.

Q31. Identify the strongest positive and negative relationships.

Q32. Check whether any variables are redundant or nearly constant.

Part 8 — Outlier & Logic Checks

Q33. Use the IQR method to identify outliers in MonthlyIncome and DistanceFromHome.

Q34. Check whether Age contains unrealistic values.

Q35. Check whether YearsAtCompany can logically exceed TotalWorkingYears.

Q36. Investigate other combinations where one variable should logically relate to another.

Part 9 — Cleaning Plan

Q37. Decide how you would handle missing numerical values.

Q38. Decide how you would handle missing categorical values.

Q39. Decide how you would handle duplicate records.

Q40. Decide how you would handle inconsistent categorical formatting.

Q41. Decide what to do with invalid values and outliers. Give a reason for every decision.

Q42. Identify columns that could potentially be dropped because they are constant, redundant, or identifiers.

Part 10 — Final EDA Report

Q43. Write 10 important observations from your EDA.

Q44. Write 5 data-quality problems you discovered and how you would fix them.

Q45. Identify the strongest observed factors associated with employee attrition. Clearly distinguish association from causation.

Q46. Write a short business summary: What should an HR team investigate further based on your EDA?

Bonus Challenge

Q47. Build a single dashboard-style figure containing 4–6 important charts.

Q48. Create an attrition profile for a typical high-risk group using only evidence from your analysis.

Q49. Repeat the most important analyses after cleaning the dataset and compare the results with the uncleaned version.